
A Complete Tutorial on Polygenic Risk Scores (PRS)

This tutorial provides a comprehensive guide to understanding, calculating, evaluating, and applying Polygenic Risk Scores (PRS). It covers the fundamental concepts, popular software tools, practical implementation, and advanced topics in the field.

Section 1: Introduction to Polygenic Risk Scores

1. What are polygenic risk scores (PRS)? Why are they important in genomics?

A **Polygenic Risk Score (PRS)** is a single value that estimates an individual's genetic predisposition for a particular trait or disease. It is calculated by adding up the effects of many genetic variants (usually single nucleotide polymorphisms, or SNPs) across the genome. Each variant contributes a small amount to the overall risk, and the PRS aggregates these small effects into a comprehensive score.

Importance in Genomics:

- **Risk Prediction:** PRS can stratify individuals into different risk categories for complex diseases like heart disease, type 2 diabetes, and breast cancer. This allows for targeted screening and preventive strategies for high-risk individuals.
- **Understanding Genetic Architecture:** PRS helps researchers understand the genetic basis of complex traits that are influenced by thousands of genes, rather than a single gene.
- **Research Tool:** PRS can be used as a variable in epidemiological studies to investigate the interplay between genetic predisposition and environmental factors (Gene-Environment interactions).
- **Validating GWAS Findings:** The predictive power of a PRS in an independent dataset serves as a validation for the genetic associations discovered in a Genome-Wide Association Study (GWAS).

2. Why use polygenic risk scores over GWAS summary statistics for trait prediction?

Genome-Wide Association Study (GWAS) summary statistics and Polygenic Risk Scores (PRS) serve different primary purposes:

- **GWAS Summary Statistics:** The main goal of a GWAS is **discovery**. It tests millions of individual genetic variants to identify which ones are statistically associated with a trait or disease. The output is a list of significant "hits" (variants) with their effect sizes and p-values. It tells you *which* genetic ingredients are important.
- **Polygenic Risk Scores (PRS):** The main goal of a PRS is **prediction**. It takes the information from GWAS summary statistics (not just the significant hits, but often hundreds or thousands of variants) and uses it to calculate a risk score for an *individual*. A PRS aggregates the effects of all these variants according to an individual's personal genotype. It uses the "recipe" from the GWAS to score an individual's overall genetic liability.

In short, you use GWAS summary statistics to find disease-associated variants, and you use a PRS to apply that knowledge to predict an individual's risk based on their unique genetic makeup.

3. How can I calculate polygenic risk scores using tools like PRSice-2, PLINK, or LDpred? What are their specific functionalities?

You can calculate PRS using several software tools, each with different strengths.

- **PLINK:** A versatile and fundamental toolkit for genetic data analysis. Its `--score` function implements the basic "clumping and thresholding" (C+T) method. It's fast and simple but may not be the most accurate, as it doesn't explicitly model Linkage Disequilibrium (LD).
 - **Functionality:** Calculates a simple weighted sum of allele counts. It's excellent for quick, straightforward PRS calculations.
- **PRSice-2:** A user-friendly and powerful R/bash script that automates the C+T method. It automatically performs clumping and tests PRS across a range of p-value thresholds to find the "best-fit" PRS. It also generates publication-quality plots.
 - **Functionality:** Best for implementing the C+T method efficiently, optimizing the p-value threshold, and evaluating PRS performance. It's a great all-in-one tool for standard PRS analysis.
- **LDpred2:** A more sophisticated Bayesian method that improves upon the C+T approach. It models the effects of all variants simultaneously, explicitly accounting for the correlational structure (LD) between them using an external LD reference panel. This generally leads to more accurate PRS.
 - **Functionality:** Best for maximizing predictive accuracy. It estimates what the "true" effect of each variant would be if all variants were independent, resulting in a more robust score. It is more computationally intensive than C+T.

4. How to perform quality control for genetic data before calculating polygenic risk scores? What are typical QC steps?

Quality control (QC) is a critical first step to ensure your PRS results are accurate and reliable. Poor-quality data can introduce noise and lead to false conclusions. The typical QC pipeline involves filtering both individuals and genetic variants.

Typical QC Steps:

1. **Individual-Level QC:**
 - **Missingness Rate:** Remove individuals with a high rate of missing genotypes (e.g., >2-5%). High missingness can indicate poor DNA quality or a failed

- genotyping experiment.
- **Sex Check:** Verify that the genetically determined sex matches the recorded sex. Mismatches can indicate sample mix-ups.
- **Relatedness:** Identify and remove related individuals (e.g., one from each pair of close relatives). Including relatives violates the assumption of independence in many statistical models.
- **Ancestry Outliers:** Use Principal Component Analysis (PCA) to identify and remove individuals whose genetic ancestry is different from the bulk of the study population.

2. Variant-level QC:

- **Missingness Rate:** Remove variants (SNPs) with a high missingness rate across all individuals (e.g., >2-5%). These are unreliable variants.
- **Hardy-Weinberg Equilibrium (HWE):** In controls, filter out variants that deviate significantly from HWE (e.g., $p < 1e-6$). Deviation can suggest genotyping errors.
- **Minor Allele Frequency (MAF):** Remove variants with a very low MAF (e.g., <1%). Rare variants have low statistical power and can be prone to genotyping errors.

After these steps, your genotype data is considered "clean" and ready for PRS calculation.

5. What are the limitations and challenges of using PRS in diverse populations?

Using PRS across different ancestral populations is one of the biggest challenges in the field, with significant scientific and ethical implications.

- **The Portability Problem:** Most GWAS have been conducted in individuals of European ancestry. PRS developed from these GWAS perform well in European-ancestry individuals but show significantly lower predictive accuracy when applied to individuals of African, Asian, or other ancestries.
- **Reasons for Poor Portability:**
 - **Differences in Allele Frequencies:** The frequency of risk variants can differ substantially across populations.
 - **Differences in Linkage Disequilibrium (LD):** The correlation patterns between variants are population-specific. A SNP that is a good proxy for a causal variant in one population may not be in another.
 - **Different Genetic Effects:** The effect size of a variant (its contribution to risk) may genuinely differ across populations due to interactions with other genes or the environment (GxE).
- **Population Stratification:** This is a major source of confounding. If a PRS is naively applied to a mixed-ancestry cohort, it may inadvertently capture ancestry differences rather than true disease risk, leading to spurious results.
- **Mitigation Strategies:**
 - **Principal Component Analysis (PCA):** The standard approach is to calculate principal components (PCs) from the genetic data and include them as covariates in the evaluation model ($\text{phenotype} \sim \text{PRS} + \text{PC1} + \dots + \text{PC10}$). This adjusts for broad-scale ancestry differences.
 - **Ancestry-Specific PRS:** The best practice is to use a GWAS conducted in an ancestry group that matches your target population.
 - **Trans-ethnic Methods:** Advanced tools like **PRS-CSx** are designed to leverage GWAS data from multiple ancestries to build a more portable and accurate PRS.
- **Ethical Implications:** Relying on European-centric PRS in clinical settings could exacerbate health disparities. It might lead to under-diagnosis in non-European populations and create a false sense of security or alarm, undermining trust in genomic medicine.

Section 2: Overview of Available Tools

6. What are the most commonly used tools for polygenic risk score analysis?

The field has a variety of tools, each suited for different methods and scenarios. The most common are:

- **PRSice-2:** A very popular, all-in-one tool for performing the clumping and thresholding (C+T) method. It's known for its speed, ease of use, and automated plotting features.
- **PLINK:** The foundational toolkit for genetic analysis. Its `--score` command is a direct and fast way to calculate a basic PRS if you have already prepared a list of SNPs and their weights.
- **LDpred2:** A leading Bayesian method that accounts for LD to improve predictive accuracy. It's an R package often used within the `bigsnpr` ecosystem. It is considered a state-of-the-art method for single-ancestry PRS.
- **lassosum:** An R package that applies penalized regression methods (LASSO and Elastic Net) to GWAS summary statistics, which also accounts for LD and can improve prediction.
- **SBayesR (in GCTB):** A Bayesian method that assumes a mixture of distributions for SNP effects. It's implemented in the GCTB command-line tool and is known for its high accuracy.
- **PRS-CSx:** A sophisticated Bayesian method specifically designed for trans-ethnic PRS, leveraging information across multiple populations to improve portability.

7. How does PRSice differ from LDpred and lassosum?

These three tools represent different classes of PRS methods.

Feature	PRSice-2 (Clumping + Thresholding)	LDpred2 (Bayesian)	lassosum (Penalized Regression)
Core Method	Selects independent SNPs based on a p-value threshold and then calculates a simple weighted sum.	Models all SNPs simultaneously, using Bayesian statistics to estimate the "true" effect of each SNP after accounting for LD.	Uses LASSO/Elastic Net regression to shrink the effects of uninformative SNPs towards zero, effectively selecting the most predictive SNPs.
	Indirectly, by "clumping"		

LD Handling	PRSice-2 (Clumping + Thresholding) (keeping only the most significant SNP in an LD block and discarding the rest).	LDpred2 (Bayesian) Directly, by using an external LD reference panel to model correlations between all SNPs.	lassosum (Penalized Regression) Directly, by using an external LD reference panel.
Accuracy	Good baseline, but often less accurate than LD-aware methods.	Generally considered one of the most accurate methods.	High accuracy, often comparable to LDpred2.
Ease of Use	Very easy. Highly automated.	Moderately complex. Requires careful preparation of summary statistics and the LD reference.	Moderately complex. Requires familiarity with R and the <code>lassosum</code> pipeline.
Best For	Quick analyses, generating initial results, and learning PRS concepts.	Maximizing predictive accuracy when a well-matched LD reference is available.	Researchers comfortable with regression-based models who want high accuracy.

8. What is the difference between individual-level data tools and summary statistic-based tools?

This distinction refers to the data required to *build* the PRS model. All methods require individual-level genotype data for the *target* samples you want to score.

- Summary Statistic-based Tools (Most common):** These tools, including **PRSice-2**, **LDpred2**, and **lassosum**, build the PRS model using only publicly available GWAS summary statistics (effect sizes, p-values) and an LD reference panel. They do not need access to the individual-level data from the original GWAS. This is a huge advantage, as individual-level data is often private and protected.
- Individual-Level Data Tools:** These tools require access to the full individual-level genotype and phenotype data from a training set to build the model. This allows for more complex modeling (e.g., machine learning models like XGBoost) but is limited by data access restrictions.

Tool Type	Data Required to Build Model	Convenience	User-Interface	Typical Tools
Summary Statistic-based	GWAS Summary Statistics	High (data is often public)	Command-line / R	PRSice-2, LDpred2, lassosum, SBayesR
Individual-Level	Individual Genotype Data	Low (data is often private)	Python / R libraries	<code>scikit-learn</code> (for ML models)

9. Which tools allow for linkage disequilibrium (LD) adjustment, and how do they do it?

Adjusting for LD is key to building an accurate PRS. Different tools do this in different ways.

- Clumping and Thresholding (C+T):** This is a simple and intuitive method for LD adjustment.
 - How it works:** It scans the genome and, for each region of correlated SNPs (an LD block), it keeps only the single most significant SNP (lowest p-value) and discards the rest. This creates a set of approximately independent SNPs.
 - Tools:** **PRSice-2** and **PLINK** use this method.
- Bayesian LD Modeling:** These methods use an external LD reference panel (e.g., from 1000 Genomes) to mathematically model the correlations between all SNPs simultaneously.
 - How it works:** They estimate the posterior mean effect of each SNP conditional on the effects of all other SNPs in the model. This "unravels" the effects of correlated SNPs, leading to a more accurate weighting.
 - Tools:** **LDpred2**, **SBayesR**, and **PRS-CSx** use this approach.
- Penalized Regression:** This method also uses an LD reference panel within a regression framework.
 - How it works:** The model automatically shrinks the effect sizes of redundant or uninformative SNPs, effectively selecting the best predictors while accounting for their correlations.
 - Tool:** **lassosum** uses this method.

In general, Bayesian and penalized regression methods provide more sophisticated LD adjustment and often yield higher predictive accuracy than C+T.

10. Are there any tools designed for ancestry-specific or trans-ethnic PRS calculation?

Yes, addressing the challenge of ancestry in PRS is a major area of research, and specialized tools have been developed.

- PRS-CSx:** This is the leading tool specifically designed for **trans-ethnic PRS**. It extends the PRS-CS model to jointly analyze GWAS summary statistics from multiple populations. It can "borrow" information across ancestries to boost predictive power, especially for underrepresented populations.
- General Tools Applied in a Trans-ethnic Context:** While not designed exclusively for it, general-purpose tools can be used to create ancestry-specific or multi-ethnic PRS.
 - LDpred2 / lassosum / PRSice-2:** You can use these tools to build an ancestry-specific PRS by providing GWAS summary statistics and an LD reference panel that are well-matched to your target ancestry (e.g., using an African-ancestry GWAS and the 1000 Genomes AFR reference panel to score individuals of African ancestry).
 - Meta-analysis approach:** One simpler strategy for trans-ethnic PRS is to calculate separate PRS for each ancestry (e.g., a European PRS and an East Asian PRS) and then combine them, for instance by taking the average or through a more complex weighting scheme.

Section 3: Installing PRS Tools

11. How do I install PRSice-2 on a Linux system using the command line?

PRSice-2 consists of an R script (`PRSice.R`) and a binary executable (`PRSice_linux`). You also need R and a few R packages installed.

1. **Install Dependencies (R and required packages):**

```
# Install R (example for Debian/Ubuntu)
sudo apt-get update
sudo apt-get install r-base r-base-dev

# Open R and install packages
R
```

Inside R:

```
install.packages(c("ggplot2", "data.table", "optparse", "pROC"))
q() # exit R
```

2. **Download PRSice-2:**

```
# Clone the repository from GitHub
git clone https://github.com/choishingwan/PRSice.git
```

3. **Make the Binary Executable:** Navigate into the cloned directory. The binary `PRSice_linux` should already be executable. If not, make it executable:

```
cd PRSice/
chmod +x PRSice_linux
```

4. **Add to PATH (Recommended):** To run PRSice-2 from any directory, move the binary and the R script to a location in your system's `$PATH` . A common place is `~/bin` or `/usr/local/bin` .

```
# Create a bin directory in your home folder if it doesn't exist
mkdir -p ~/bin

# Copy the files
cp PRSice.R ~/bin/
cp PRSice_linux ~/bin/

# Add ~/bin to your PATH (add this line to your ~/.bashrc or ~/.zshrc)
export PATH="$HOME/bin:$PATH"

# Apply the change to your current terminal session
source ~/.bashrc
```

Now you can run PRSice-2 from anywhere.

12. What dependencies or software are needed to run PRSice-2?

As outlined above, the key dependencies for PRSice-2 are:

1. **R:** The statistical programming language.
2. **R Packages:**
 - `optparse` : For parsing command-line arguments.
 - `data.table` : For fast data manipulation.
 - `ggplot2` : For generating plots (e.g., the bar chart of PRS performance).
 - `pROC` : For calculating AUC for binary traits.
3. **PRSice-2 Binary:** The pre-compiled executable for your operating system (`PRSice_linux` , `PRSice_mac` , or `PRSice_win.exe`).
4. **(Optional) PLINK:** PRSice-2 can call PLINK for clumping, so it's good to have PLINK 1.9 installed and in your `PATH` .

13. How can I install and configure LDpred2 in R?

LDpred2 is an R package available from Bioconductor. It's best used within the `bigsnpr` ecosystem, which helps manage large genetic datasets efficiently.

1. **Install `bigsnpr` and `ldpred2`:** Open R and run the following commands.

```
# Install the 'remotes' package if you don't have it
if (!require("remotes")) install.packages("remotes")

# Install bigsnpr from its GitHub repository
remotes::install_github("privefl/bigsnp")

# The latest version of bigsnpr includes ldpred2 functionality
```

2. **Configuration:** The main "configuration" for LDpred2 involves preparing your data correctly:
 - **GWAS Summary Statistics:** Must be harmonized so that the effect allele (A1) matches the allele in your LD reference panel. The `bigsnp::snp_match()` function is essential for this.
 - **LD Reference Panel:** You need genotype data for a reference population (e.g., 1000 Genomes) in PLINK format (`.bed` / `.bim` / `.fam`). You will also need to compute an LD correlation matrix from this panel, which can be done once and saved as a `.rds` file for future use with `bigsnp` .

14. How do I install lassosum in R, and what libraries does it require?

`lassosum` is an R package that can be installed directly from CRAN.

1. **Install `lassosum`** : Open R and run:

```
install.packages("lassosum")
```

2. **Load the Library:**

```
library(lassosum)
```

3. **Dependencies:** Installing `lassosum` will automatically install its required dependencies, which include `data.table` for data handling and `pgenlibr` for reading PLINK 2 files. Like LDpred2, `lassosum` requires properly formatted summary statistics and an LD reference panel in PLINK format.

15. What is SBayesR and how do I install GCTB to run it?

SBayesR is a Bayesian PRS method that assumes SNP effects come from a mixture of four normal distributions. It is not a standalone tool but is implemented within the **GCTB (Genome-wide Complex Trait Bayesian analysis)** software package.

To install GCTB:

1. **Clone the Repository:** GCTB is available on GitHub. You need `git` and a C++ compiler (`g++`).

```
git clone https://github.com/gctb/gctb.git
```

2. **Compile the Software:** Navigate into the directory and use `make` .

```
cd gctb/
make
```

3. **Add to PATH:** After successful compilation, an executable named `gctb` will be created. You should move this to a directory in your `$PATH` for easy access.

```
# Example: Move to ~/bin
mkdir -p ~/bin
cp gctb ~/bin/
# Ensure ~/bin is in your PATH (see step 11.4)
```

Section 4: Running PRS Tools

16. What are the required input files for PRSice-2, and how should they be formatted?

PRSice-2 requires at least two files, and usually a few more for a full analysis.

1. **Base Data (GWAS Summary Statistics):**
 - **Purpose:** Provides the SNP effect sizes (BETA or OR) and p-values to build the PRS model.
 - **Format:** A plain text file with a header. Must contain columns for:
 - SNP identifier (e.g., `SNP`)
 - Effect allele (e.g., `A1`)
 - Non-effect allele (e.g., `A2`)
 - P-value (e.g., `P`)
 - Effect size (e.g., `BETA` or `OR`)
 - Example: `gwas_summary.txt`

2. Target Data (Individual Genotypes):

- **Purpose:** The genotype data for the individuals you want to score.
- **Format:** PLINK binary format (`.bed` , `.bim` , `.fam`).
- Example: `my_genotypes.bed` , `my_genotypes.bim` , `my_genotypes.fam`

3. Phenotype File (for evaluation):

- **Purpose:** To evaluate how well the calculated PRS predicts the trait in your target data.
- **Format:** A plain text file with a header. Must contain:
 - Family ID (`FID`)
 - Individual ID (`IID`)
 - The phenotype value (e.g., `Height` for quantitative, or `Disease` with 1=case, 0=control for binary).
- Example: `phenotypes.txt`

4. Covariate File (for adjustment):

- **Purpose:** To adjust for potential confounders like age, sex, and genetic principal components.
- **Format:** Same format as the phenotype file (`FID` , `IID` , followed by columns for each covariate like `Age` , `Sex` , `PC1` , `PC2` , etc.).

17. Provide a sample command to run PRSice-2 using PLINK genotype data and GWAS summary stats.

This command runs a full PRSice-2 analysis for a binary disease outcome, including evaluation and adjustment for covariates.

```
Rscript PRSice.R --prsice PRSice_linux \  
  --base gwas_summary.txt \  
  --target my_genotypes \  
  --pheno phenotypes.txt \  
  --cov covariates.txt \  
  --binary-target T \  
  --stat BETA \  
  --beta \  
  --thread 8 \  
  --out my_prs_analysis
```

Command Breakdown:

- `Rscript PRSice.R` : Executes the PRSice R script.
- `--prsice PRSice_linux` : Specifies the path to the PRSice binary.
- `--base gwas_summary.txt` : Path to the GWAS summary statistics file.
- `--target my_genotypes` : The prefix for your PLINK target data (`.bed` / `.bim` / `.fam`).
- `--pheno phenotypes.txt` : The phenotype file.
- `--cov covariates.txt` : The covariate file.
- `--binary-target T` : Specifies the phenotype is binary (True). Use `F` for quantitative.
- `--stat BETA` : Tells PRSice the effect size column in `--base` is named `BETA` .
- `--beta` : Confirms the effect sizes are BETAs (log odds ratios), not ORs.
- `--thread 8` : Use 8 CPU threads to speed up the analysis.
- `--out my_prs_analysis` : The prefix for all output files.

18. How do I prepare genotype and LD reference files for LDpred2?

Preparing data for LDpred2 requires two main components and careful harmonization.

1. GWAS Summary Statistics (`sumstats`):

- This is a data frame in R. It must contain columns for SNP ID (`rsid`), chromosome (`chr`), position (`pos`), effect allele (`a1`), non-effect allele (`a0`), effect size (`beta`), standard error (`beta_se`), and sample size (`n_eff`).
- **Crucially, this file must be harmonized with the LD reference.** This means the `rsid` and alleles (`a1` , `a0`) must match those in the LD reference panel. The `bigsnpr::snp_match()` function is designed for this step.

2. LD Reference Panel (`LD_ref`):

- This starts as a PLINK file set (`.bed` / `.bim` / `.fam`) of an external reference population (e.g., 1000 Genomes or UK Biobank).
- Using the `bigsnpr` package, you read this data into a special `bigSNP` object.
- From this object, you calculate a correlation matrix. For efficiency, this is usually done once and the result is saved as an `.rds` file. This `.rds` file containing the correlation matrix and related information is what LDpred2 directly uses.

19. Provide a complete R script to run LDpred2 with summary statistics and reference data.

This script uses the `bigsnpr` pipeline, which is the standard way to run LDpred2.

```

# Load libraries
library(bigsnp)
library(ggplot2)

# --- 1. SETUP ---
# Attach the bigSNP object of the LD reference panel (e.g., 1000 Genomes)
# This assumes you have already created 'ref.rds' from PLINK files
obj.bigSNP <- snp_attach("data/ref.rds")
G <- obj.bigSNP$genotypes
map <- obj.bigSNP$map[,c("marker.ID", "chromosome", "physical.pos", "allele1", "allele2")]
names(map) <- c("rsid", "chr", "pos", "a0", "a1")

# Read in your GWAS summary statistics
sumstats <- bigreadr::fread2("data/gwas_summary.txt")

# --- 2. HARMONIZE DATA ---
# Match summary statistics with the LD reference panel
# This resolves strand flips and mismatches, which is a critical step
info_snp <- snp_match(sumstats, map)

# --- 3. CALCULATE LD MATRIX ---
# This is done once and saved for reuse
# If the correlation matrix is not pre-computed:
# for (chr in 1:22) {
#   ind.chr <- which(info_snp$chr == chr)
#   corr_file <- paste0("data/LD_chr", chr, ".rds")
#   snp_cor(G, ind.col = info_snp$`_NUM_ID_`[ind.chr], ncores = 4,
#           infos.pos = info_snp$pos[ind.chr], size = 3 / 1000,
#           outfile = corr_file)
# }
# Here, we assume it's already done and we load it
tmp <- tempfile(tmpdir = "data/tmp-data")
ld_mat <- snp_ld_scores(G, info_snp$`_NUM_ID_`, ncores = 4)
ld <- info_snp$ld <- ld_mat

# --- 4. RUN LDPRED2 ---
# Run the LDpred2-grid model (tests a range of heritability and sparsity parameters)
# This is a robust approach if you don't know the trait's genetic architecture
beta_grid <- snp_ldpred2_grid(
  corr = snp_attach("data/LD_chr_all.rds")$corr, # Load the pre-computed LD matrix
  df_beta = info_snp,
  n_cores = 4
)

# --- 5. CALCULATE PRS IN TARGET DATA ---
# Attach target genotype data
target.bigSNP <- snp_attach("data/target_data.rds")

# Calculate the PRS using the betas from LDpred2
final_prs <- big_prodMat(
  target.bigSNP$genotypes,
  beta_grid,
  ind.col = info_snp$`_NUM_ID_`
)

```

20. What is the format of input summary statistics for lassosum, and how is the LD reference provided?

- Input Summary Statistics:** `lassosum` requires the GWAS summary statistics to have SNP IDs (e.g., rsIDs) and correlation values (`cor`). The correlation is usually derived from the BETA and p-value: $cor = beta / \sqrt{n * se^2 + beta^2}$. The `lassosum` functions can often derive this for you if you provide BETA, SE, and N. A column for the effect allele (`A1`) is also required for harmonization.
- LD Reference:** The LD reference is provided as a PLINK file set (`.bed` / `.bim` / `.fam`). You simply provide the file path prefix to the `lassosum.pipeline` function. The tool handles the LD calculation internally.

21. How do I run lassosum using R commands and visualize the output?

Here is a sample workflow for `lassosum`.

```
# Load library
library(lassosum)

# --- 1. PREPARE DATA ---
# Read in GWAS summary statistics and convert BETA to correlation (cor)
ss <- data.table::fread("data/gwas_summary.txt")
cor <- p2cor(p = ss$P, n = ss$N, sign = ss$BETA)

# Specify path to LD reference PLINK files
ld.bfile <- "data/ref_1000G"

# --- 2. RUN LASSOSUM PIPELINE ---
# This function handles the entire process
out <- lassosum.pipeline(
  cor = cor,
  chr = ss$CHR,
  pos = ss$POS,
  A1 = ss$A1,
  A2 = ss$A2,
  ref.bfile = ld.bfile,
  test.bfile = "data/target_data" # Your target PLINK data
)

# --- 3. VALIDATE AND GET BEST PRS ---
# The 'out' object contains results for a range of tuning parameters.
# We validate them against a phenotype to find the best one.
# 'v' contains the validation results (e.g., R-squared)
v <- validate(out, pheno = target.pheno, covar = target.covar)

# The best PRS is stored in v$best.pgs
best_prs <- v$best.pgs

# --- 4. VISUALIZE ---
# Plot R-squared against the tuning parameter lambda
plot(v)

# Create a scatter plot of phenotype vs PRS
plot(best_prs, target.pheno, xlab="Polygenic Score", ylab="Phenotype")
abline(a=0, b=1, col="red") # Add a reference line
```

22. What is the full command-line pipeline to run SBayesR using GCTB on summary stats and LD matrices?

SBayesR is run using the `gctb` command-line tool. It requires GWAS summary statistics in a specific `.ma` format and an LD matrix.

Step 1: Format GWAS Summary Statistics Convert your standard GWAS summary file to the GCTB `.ma` format, which requires `SNP`, `A1`, `A2`, `freq`, `b`, `se`, `p`, `N`.

Step 2: Generate LD Matrix (if you don't have one) GCTB can generate the required LD matrix from a PLINK file set.

```
gctb --bfile ref_1000G --make-mldm --out ref_ldm
```

This creates `ref_ldm.mldm`, a multi-LD-matrix file.

Step 3: Run SBayesR Now, run the SBayesR analysis.


```
gctb --sbayes R \
  --ldm ref_ldm.mldm \
  --gwas-summary my_gwas.ma \
  --chain-length 2000 \
  --burn-in 1000 \
  --out my_sbayesr_analysis
```

- `--sbayes R` : Specifies the SBayesR method.
- `--ldm ref_ldm.mldm` : Provides the LD matrix file.
- `--gwas-summary my_gwas.ma` : Provides the formatted GWAS summary stats.
- `--chain-length` / `--burn-in` : MCMC parameters for the Bayesian model.
- `--out` : Specifies the prefix for output files.

The main output is `my_sbayesr_analysis.snpRes`, which contains the estimated posterior effect size for each SNP, ready to be used for scoring.

23. How can I compute PRS using PLINK's `--score` function, and what is the required file format?

PLINK's `--score` function provides a direct and fast way to calculate a PRS.

Required Files:

1. **Genotype Data:** Your target data in PLINK format (`.bed` / `.bim` / `.fam`).
2. **Score File:** A plain text file with SNP weights. The format is flexible, but a simple version is:

```
# scorefile.txt
rs1234 A 0.05
rs5678 G -0.02
rs9101 T 0.01
...
```

- **Column 1:** SNP ID (must match the `.bim` file).
- **Column 2:** Effect allele.
- **Column 3:** Weight (effect size/BETA).

PLINK Command:

```
plink --bfile my_target_data \
  --score scorefile.txt 1 2 3 header \
  --out my_plink_prs
```

Command Breakdown:

- `--bfile my_target_data` : The PLINK file set for your target individuals.
- `--score scorefile.txt 1 2 3 header` : Specifies the score file and its structure.
 - `1` : The SNP ID is in column 1.
 - `2` : The effect allele is in column 2.
 - `3` : The weight is in column 3.
 - `header` : Tells PLINK to ignore the first line of the score file.
- `--out my_plink_prs` : The prefix for the output file.

The primary output is `my_plink_prs.profile`, a file containing the `FID`, `IID`, and calculated PRS for each individual.

Section 5: Evaluating and Interpreting PRS

24. What metrics can be used to evaluate the performance of a PRS model (e.g., AUC, R²)?

The metric you use depends on whether your phenotype is quantitative or binary. Evaluation is typically done by fitting a regression model in the target dataset.

For Quantitative Traits (e.g., height, BMI):

- **Metric:** R-squared (R²)
- **Interpretation:** The proportion of the variance in the phenotype that is explained by the PRS.
- **How to calculate:** Fit a linear regression model: `phenotype ~ PRS + covariates`. The (incremental) R² of the model tells you the predictive power of the PRS after accounting for covariates.

For Binary Traits (e.g., disease case/control):

- **Metric 1: Area Under the ROC Curve (AUC)**
 - **Interpretation:** A measure of discrimination. An AUC of 0.5 means the PRS is no better than random guessing, while an AUC of 1.0 means perfect prediction.

- **How to calculate:** Fit a logistic regression model: `phenotype ~ PRS + covariates` . Then use the predicted probabilities from this model to calculate the AUC.
- **Metric 2: Nagelkerke's R² (Pseudo-R²)**
 - **Interpretation:** An analogue to R² for logistic regression. It represents the improvement in the model fit provided by the PRS.
- **Metric 3: Odds Ratio (OR) per Standard Deviation**
 - **Interpretation:** The change in disease odds for every one standard deviation increase in the PRS. This is a highly interpretable measure of effect size. For example, an OR of 1.5 means that a 1 SD increase in PRS is associated with a 50% increase in the odds of having the disease.

25. How can covariates such as age, sex, or principal components be incorporated into PRS analysis?

Covariates are not typically used in the *calculation* of the PRS itself. Instead, they are incorporated during the **evaluation** step to get an unbiased estimate of the PRS's predictive power.

The standard procedure is a two-step process:

1. **Step 1: Calculate the PRS.** Use a tool like PRSice-2, LDpred2, or PLINK to calculate a single PRS value for each individual in your target dataset.
2. **Step 2: Run a Regression Model.** Use statistical software (like R) to fit a regression model where the phenotype is the outcome and the PRS and covariates are predictors.

- **For a quantitative trait:**

```
# R code
model <- lm(phenotype ~ PRS + Age + Sex + PC1 + PC2 + PC3 + PC4, data = my_data)
summary(model)
```

- **For a binary trait:**

```
# R code
model <- glm(disease_status ~ PRS + Age + Sex + PC1 + PC2 + PC3 + PC4, family = "binomial", data = my_data)
summary(model)
```

By including covariates in the model, you are estimating the effect of the PRS **independent** of the effects of age, sex, and genetic ancestry (as captured by the PCs). This is the correct way to assess the unique contribution of the polygenic score.

26. How do you compare PRS models across tools using the same phenotype and dataset?

To perform a "bake-off" and compare the performance of PRS from different methods (e.g., PRSice-2 vs. LDpred2 vs. SBayesR), you must follow a rigorous and fair procedure.

1. **Use Identical Data:** The comparison must be performed on the exact same set of individuals (target genotype data, phenotypes, and covariates).
2. **Generate PRS from Each Method:**
 - Run PRSice-2 to generate `PRS_C+T` .
 - Run LDpred2 to generate `PRS_LDpred2` .
 - Run SBayesR to generate `PRS_SBayesR` .
 - ...and so on for any other methods.
3. **Standardize the PRS:** Before comparing, it's best practice to standardize each PRS (scale to a mean of 0 and a standard deviation of 1). This puts them on a common scale, making their regression coefficients more comparable.

```
my_data$PRS_C_T_scaled <- scale(my_data$PRS_C_T)
my_data$PRS_LDpred2_scaled <- scale(my_data$PRS_LDpred2)
```

4. **Fit Identical Regression Models:** For each standardized PRS, fit the exact same regression model and extract the performance metric.
 - **Model 1 (C+T):** `lm(phenotype ~ PRS_C_T_scaled + covariates) -> Get R21`
 - **Model 2 (LDpred2):** `lm(phenotype ~ PRS_LDpred2_scaled + covariates) -> Get R22`
 - **Model 3 (SBayesR):** `lm(phenotype ~ PRS_SBayesR_scaled + covariates) -> Get R23`
5. **Compare Metrics:** The method that yields the highest performance metric (e.g., the highest R² or AUC) is considered the best-performing model for that specific trait and dataset. You can also use formal statistical tests (e.g., DeLong's test for comparing AUCs) to see if the differences in performance are statistically significant.

27. What is cross-validation in PRS evaluation, and how can it be applied?

Cross-validation (CV) is a powerful statistical technique used to get a more robust and honest estimate of a model's predictive performance. It helps prevent **overfitting**, which occurs when a model performs well on the data it was trained on but fails to generalize to new, unseen data.

In the context of PRS, a method like PRSice-2 or LDpred2-grid has "tuning parameters" (e.g., the p-value threshold in C+T). CV is used to select the optimal tuning parameter.

How it works (k-fold CV):

1. **Split the Data:** Divide your target dataset randomly into k equal-sized parts (or "folds"). A common choice is $k=5$ or $k=10$.
2. **Iterate:** Repeat the following process k times:
 - Hold out one fold as the **testing set**.
 - Use the remaining $k-1$ folds as the **training set**.
 - On the training set, build your PRS models across a range of different tuning parameters (e.g., p-value thresholds from $5e-8$ to 1.0).
 - Apply these trained models to the testing set and calculate the performance (e.g., R^2) for each tuning parameter.
3. **Average and Select:** After repeating this for all k folds, you average the performance metric across the folds for each tuning parameter. The tuning parameter that gives the best average performance is chosen as the optimal one.
4. **Final Model:** Finally, you re-train your PRS model on the **entire dataset** using the optimal tuning parameter you just found. The performance of this final model is your robust, cross-validated estimate of its predictive power.

Using CV gives you more confidence that your PRS model will perform well on new individuals, not just the ones in your sample.

Section 6: Advanced Use Cases and Integrations

28. How can PRS be integrated into clinical risk prediction models?

Clinical risk models (e.g., the Framingham Risk Score for cardiovascular disease) traditionally use non-genetic factors like age, sex, blood pressure, cholesterol, and smoking status to predict disease risk. A PRS can be integrated to see if genetic information provides additional predictive value.

The process is as follows:

1. **Establish a Baseline Model:** Create a clinical risk model using only traditional risk factors.
 - `Model_Clinical = glm(disease ~ Age + Sex + Smoking + Cholesterol, family="binomial")`
2. **Create an Integrated Model:** Add the PRS as a new predictor to the baseline model.
 - `Model_Integrated = glm(disease ~ Age + Sex + Smoking + Cholesterol + PRS, family="binomial")`
3. **Evaluate the Improvement:** Compare the performance of `Model_Integrated` to `Model_Clinical`.
 - **AUC Improvement:** Does the AUC of the integrated model significantly increase compared to the clinical model? (Use DeLong's test).
 - **Net Reclassification Improvement (NRI):** How many individuals are correctly re-classified into higher or lower risk categories after adding the PRS? This is a key metric for clinical utility.

If adding the PRS significantly improves prediction and reclassification, it demonstrates its potential clinical utility for refining risk estimates.

29. What is ancestry-specific PRS, and how is it computed using tools like PRS-CSx?

- **Ancestry-Specific PRS:** This is a PRS built and applied within a single, specific ancestral population. For example, to create a PRS for individuals of East Asian ancestry, you would ideally use:
 1. A large GWAS conducted in an East Asian cohort.
 2. An LD reference panel from the same ancestry (e.g., 1000 Genomes EAS). This approach maximizes accuracy by ensuring the genetic architecture (allele frequencies, LD patterns) is consistent between the GWAS, the LD reference, and the target samples.
- **Trans-ethnic PRS with PRS-CSx:** PRS-CSx is a specialized tool that goes beyond single-ancestry analysis. It is designed to compute PRS by leveraging information from **multiple** ancestries simultaneously.
 - **How it works:** PRS-CSx takes GWAS summary statistics from two or more populations (e.g., European and African) as input. It uses a coupled Bayesian model that assumes genetic effects are partially shared across populations but also allows for population-specific effects. By jointly modeling the data, it can "borrow" statistical power from the larger GWAS (usually European) to improve the estimation of SNP effects in the smaller, underrepresented GWAS.
 - **Benefit:** This approach can significantly improve PRS accuracy in non-European populations, helping to mitigate the portability problem and reduce health disparities.

30. How can machine learning methods like ElasticNet or XGBoost be used in PRS modeling?

While standard PRS methods are forms of linear models, more complex machine learning (ML) methods can also be applied, though this is an active area of research.

- **Elastic Net:** This is a form of penalized linear regression. It's a natural extension of PRS methods like `lassosum`.
 - **How it works:** When fitting a model with thousands of SNPs as predictors, Elastic Net automatically performs variable selection by shrinking the coefficients of uninformative SNPs to zero. It is particularly good at handling groups of correlated SNPs (LD blocks), often selecting the whole group rather than just one SNP.
 - **Application:** Can be applied to individual-level training data or, with some adaptations, to summary statistics to potentially improve upon standard methods.
- **XGBoost (Gradient Boosted Trees):** This is a powerful, non-linear ML method.
 - **How it works:** XGBoost builds an ensemble of decision trees to make predictions. Unlike linear models, it can automatically capture complex non-linear relationships and interactions between SNPs (epistasis).
 - **Application:** Requires large, individual-level training datasets. Its application in PRS is still exploratory because the results are harder to interpret ("black box" problem), and it's computationally very intensive. It holds promise for capturing parts of the genetic architecture that linear models miss.

31. How do you perform a trans-ethnic PRS analysis using multiple population references?

Performing a robust trans-ethnic PRS analysis requires a careful strategy to leverage information from multiple populations while accounting for their genetic differences.

Method 1: The Meta-Analysis Approach (Simpler)

1. **Create Ancestry-Specific PRS:** For each reference population (e.g., European, East Asian), generate a separate PRS.
 - `PRS_EUR` = a PRS trained using a European GWAS and LD reference.
 - `PRS_EAS` = a PRS trained using an East Asian GWAS and LD reference.
2. **Score Target Individuals:** Calculate both scores (`PRS_EUR` and `PRS_EAS`) for all individuals in your (potentially admixed) target cohort.
3. **Combine the Scores:** Combine the scores in a regression model to predict the phenotype.
 - `phenotype ~ PRS_EUR + PRS_EAS + covariates` This approach allows you to see the contribution of each ancestry-specific score. You can also combine them using methods like taking the average or weighting them by the individual's ancestry proportions.

Method 2: The Joint Modeling Approach (Advanced, Recommended)

1. **Use a Specialized Tool:** Employ a tool like **PRS-CSx** that is specifically designed for this purpose.
2. **Provide Multiple Inputs:** Give PRS-CSx the GWAS summary statistics from all relevant populations (e.g., `gwas_eur.txt` , `gwas_eas.txt`) and their corresponding LD reference panels.
3. **Run the Joint Model:** The tool will perform the Bayesian analysis, jointly modeling SNP effects across populations.
4. **Get a Single, Trans-ethnic PRS:** The output will be a single set of SNP weights that has been optimized by borrowing information across ancestries. This set of weights can then be used to calculate a single, more powerful and portable PRS for your target individuals.

32. How can gene-environment interactions be included in PRS models?

A gene-environment (GxE) interaction occurs when the effect of a genetic factor (like a PRS) on a trait depends on an environmental exposure. For example, a high genetic risk for heart disease might only manifest in individuals who also have a high-fat diet.

To model GxE interactions with PRS:

1. **Collect Data:** Your study must have data on three things for each individual:
 - The phenotype of interest.
 - The calculated PRS.
 - The environmental exposure (e.g., smoking status, physical activity level, dietary patterns).
2. **Fit an Interaction Model:** Use a regression model that includes the PRS, the environmental factor, and an **interaction term** between them.

```
# E.g., Modeling how the effect of PRS on BMI depends on physical activity
# high_activity is a binary variable (1=high activity, 0=low)

model_gxe <- lm(BMI ~ PRS + high_activity + PRS * high_activity + covariates, data=my_data)
summary(model_gxe)
```

3. **Interpret the Interaction Term (`PRS * high_activity`):**
 - If the coefficient for this term is statistically significant, it provides evidence for a GxE interaction.
 - A **negative** coefficient would mean that the effect of PRS on BMI is weaker (blunted) in the high-activity group.
 - A **positive** coefficient would mean the effect of PRS on BMI is even stronger in the high-activity group.

Investigating GxE interactions is crucial for understanding disease mechanisms and developing personalized lifestyle recommendations. However, these analyses require very large sample sizes to have sufficient statistical power to detect the interaction effect.

Section 7: Troubleshooting and Best Practices

33. What are common issues that arise when running PRS tools and how can they be resolved?

Here are some of the most frequent errors and their solutions:

Error Type	Common Messages	Solution
File Path Errors	No such file or directory, Cannot open file	Double-check that the file names and paths are spelled correctly. Use absolute paths (<code>/home/user/data/file.txt</code>) instead of relative paths (<code>./file.txt</code>) to avoid ambiguity.
Formatting Errors	Invalid argument, Problem parsing header, Wrong number of columns	Carefully check your input files (GWAS summary stats, covariate files). Ensure they are plain text, have the correct delimiters (e.g., tabs or spaces), and that the header names match exactly what the tool expects (e.g., <code>SNP</code> , <code>BETA</code> , <code>P</code>). Remove any special characters or formatting.
Data Mismatch	SNP ID mismatch, No matching variants, All samples removed	This is often a harmonization issue. The SNP IDs (<code>rs...</code>), chromosome, position, and alleles in your GWAS summary stats must match those in your target <code>.bim</code> file and your LD reference panel. Use tools like <code>bigsnpr::snp_match</code> to align your files before running the PRS tool.

Memory Errors	Failed to allocate memory, Killed	Your analysis is using more RAM than is available. Reduce the number of threads/cores (thread), run the analysis on a high-performance computing cluster, or use tools like bigsnpr that are designed for memory-efficient handling of large datasets.
Parameter Errors	Invalid number of parameters, Unrecognized option	Read the tool's documentation carefully. Make sure you are using the correct flags (e.g., --beta vs --or, --binary-target T for case/control) and providing all required arguments.

34. How do different LD reference panels (e.g., 1000 Genomes vs UK Biobank) impact PRS results?

The choice of LD reference panel is critical for methods like LDpred2 and lassosum and can significantly impact PRS accuracy.

- **Ancestry Match is Key:** The most important factor is how well the ancestry of the LD reference panel matches the ancestry of your GWAS summary statistics. A mismatch will lead to inaccurate LD estimation and a less predictive PRS.
 - **For European-ancestry GWAS:** Using a European reference panel from **1000 Genomes (EUR)** or a subset of **UK Biobank** will produce good results. UK Biobank is much larger, which can provide more precise LD estimates, but 1000 Genomes is often sufficient and more accessible.
 - **For non-European GWAS:** It is crucial to use an ancestry-matched panel, e.g., **1000 Genomes AFR** for an African-ancestry GWAS or **1000 Genomes EAS** for an East Asian-ancestry GWAS. Using a European LD reference for a non-European GWAS is a common mistake that will severely degrade PRS performance.
- **Sample Size:** A larger reference panel (like UK Biobank, ~500k individuals) can estimate LD correlations more accurately than a smaller one (like 1000 Genomes, ~2.5k individuals), which can lead to a modest improvement in PRS accuracy.
- **SNP Density:** The reference panel should have good coverage of the SNPs present in your GWAS summary statistics. Modern imputation panels like HRC or TopMed provide excellent density.

In summary: Always prioritize a well-matched ancestry. If multiple options with the correct ancestry exist, a larger sample size is generally better.

35. What is the effect of imputation quality on PRS computation?

Genotype imputation is the process of statistically inferring un-genotyped variants. The quality of this imputation has a direct effect on PRS accuracy.

- **How it works:** Imputation algorithms provide a quality score for each imputed SNP, often called an **INFO score** or **R-squared (R²)**. This score ranges from 0 to 1 and reflects the algorithm's confidence in the imputed genotype call. An INFO score of 1 means perfect certainty; a score near 0 means the imputation is no better than guessing based on allele frequency.
- **Effect on PRS:**
 - **High-Quality Imputation (INFO > 0.8):** Well-imputed SNPs can be confidently included in a PRS calculation, increasing the number of variants used and potentially improving predictive power.
 - **Low-Quality Imputation (INFO < 0.3):** Including poorly imputed SNPs introduces significant noise and error into the PRS. The "genotypes" for these SNPs are unreliable, and their inclusion will degrade the accuracy of the final score.
- **Best Practice:** It is standard practice to **filter out** variants with low imputation quality before calculating a PRS. A common threshold is to only include SNPs with an INFO score > 0.8. This is a crucial post-imputation QC step that ensures the PRS is built from reliable genetic information.

36. How do I handle missing or low-quality SNPs during PRS generation?

Handling missing and low-quality SNPs is a core part of the pre-analysis QC pipeline.

1. **Directly Genotyped Data (e.g., from a SNP array):**
 - **QC Filtering:** Before imputation or PRS calculation, filter out low-quality SNPs based on standard QC metrics:
 - High missingness rate (e.g., `plink --geno 0.02`)
 - Deviation from Hardy-Weinberg Equilibrium (e.g., `plink --hwe 1e-6`)
 - Low Minor Allele Frequency (e.g., `plink --maf 0.01`)
2. **Imputed Data:**
 - **Imputation:** For array data, impute genotypes using a reference panel like the 1000 Genomes or HRC to increase SNP density. This "fills in" many of the missing genotypes.
 - **Post-Imputation Filtering:** After imputation, it's critical to filter the newly available SNPs based on their imputation quality.
 - **INFO Score/R²:** Only retain variants with a high imputation quality score (e.g., INFO > 0.8).

By first removing inherently low-quality variants and then using high-confidence imputed variants, you create a clean and dense set of SNPs ready for use in a PRS calculation.

37. What steps should be taken to ensure reproducibility and version control in PRS studies?

Reproducibility is the cornerstone of good science. Ensuring your PRS analysis is reproducible is essential for its credibility.

1. **Script Everything:** Do not perform any analysis steps by manually clicking buttons in a GUI. Write a script (e.g., a Bash shell script or an R script) that runs the entire pipeline from start to finish. This script is your analysis's complete recipe.
2. **Use Version Control:** Store your analysis scripts in a version control system like **Git**, and host them on a platform like **GitHub**. This tracks every change you make and allows others (or your future self) to access the exact code used to generate the results.
3. **Document Software Versions:** At the beginning of your script or in a **README** file, explicitly state the versions of all software used.

- PRSice-2 v2.3.5
- PLINK v1.90b6.21
- R v4.2.0 and the versions of key R packages (data.table v1.14.2 , bigsnpr v1.10.3 , etc.).

4. **Use Containers (Gold Standard):** For maximum reproducibility, use containerization technology like **Docker** or **Singularity**. A container packages your entire analysis environment—the operating system, software, libraries, and scripts—into a single, portable image. Anyone can download and run your container to perfectly replicate your results, regardless of their local machine's setup.
5. **Document Data Sources:** Clearly document the source and version of all your input data. Provide download links or Digital Object Identifiers (DOIs) for public data like GWAS summary statistics and LD reference panels.
6. **Set Random Seeds:** If any step in your analysis involves random processes (e.g., cross-validation splits), set a random seed at the beginning of your script (e.g., `set.seed(123)` in R). This ensures the "random" numbers are the same every time the script is run.